INSTITUTE OF ENGINEERING & MANAGEMENT

KOLKATA

COURSE TITLE: BACHELOR OF TECHNOLOGY

SUBJECT: PYTHON PROGRAMMING LABORATORY

PROJECT TITLE: WIRELESS COVERAGE MAPPER

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SECTION: B1

SUBMITTED TO: Prof. INDRANIL BASU

DEPARTMENT NAME: ELECTRONICS AND COMMUNICATION ENGINEERING

ACADEMIC SESSION: 2026-2027

DATE OF SUBMISSION: 14.11.2026

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WIRELESS COVERAGE MAPPER

Abstract

This project presents a Python-based wireless coverage analysis tool that examines recorded Wi-Fi Received Signal Strength Indicator (RSSI) measurements at points on a floor or campus. The application validates survey data, rejects incomplete or invalid records with reasons, analyses spatial measurements using NumPy and Pandas, and generates maps, histograms, grid summaries and a human-readable report. A reusable SurveyPoint data model and command-line workflow form the core of the implementation.

The repository includes a small demonstration dataset containing 12 rows. Eleven records have valid coordinates and one row is rejected because its x-coordinate is missing. The default weak-signal threshold is -67 dBm; readings below this threshold are identified as weak zones. The included data is illustrative and is not presented as a live or representative survey of a real building. The tool distinguishes measured points from grid-cell averages and leaves empty grid cells unmeasured.

Keywords

Wireless coverage; Wi-Fi; RSSI; signal strength; Python; NumPy; Pandas; Matplotlib; CSV; heatmap; spatial analysis.

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1. Introduction

Wireless connectivity depends on the strength and quality of radio signals received at different locations. Within a building or campus, signal strength may vary due to distance from the access point, walls, obstacles, interference and the surrounding environment. A structured site survey helps identify locations where the received signal is relatively weak.

Wireless Coverage Mapper is designed to analyze recorded signal-strength samples associated with two-dimensional survey coordinates. It reads a CSV file, validates each record, calculates location-wise summaries and produces visual outputs. The project focuses on recorded measurements and transparent analysis rather than claiming to simulate radio propagation or predict actual throughput.

2. Problem Statement and Objectives

Problem statement: Develop a Python application that processes Wi-Fi signal-strength readings collected at known coordinate points, identifies weak-signal observations, compares signal conditions across locations and visualises spatial patterns. The program must validate input data and clearly report rejected rows so that missing or invalid coordinates do not create misleading spatial results.

The principal objectives are:

- Read wireless survey records from a structured CSV file.
- Validate required columns, coordinate values and signal-strength readings.
- Represent each valid measurement using a SurveyPoint data model.
- Use NumPy for pairwise spatial distances and grid-based signal summaries.
- Use Pandas to compare signal strength across location identifiers.
- Identify weak zones using a configurable RSSI threshold.
- Generate a measured-point map, signal histogram, grid heatmap and machine-readable summaries.
- Export accepted points, weak zones, rejected rows and a human-readable analysis report.

3. Software and System Requirements

Component	Purpose
Python 3	Core program execution and command-line interface
CSV module	Reading structured survey measurements
NumPy	Numerical operations, pairwise distances and grid calculations
Pandas	Tabular data handling, filtering and location-wise comparison
Matplotlib	Signal map, histogram and grid heatmap visualization
pytest	Automated verification of validation and analysis functions
Ruff / CI workflow	Code formatting, linting and automated quality checks

The project runs from a terminal in an environment with Python and the dependencies listed in requirements.txt. The included workflow accepts a CSV survey file and writes generated outputs to a specified output directory.

4. System Design and Methodology

4.1 Input data and schema

The required CSV fields are location_id, x, y and signal_dbm. The optional label field describes the surveyed area. The x and y fields represent two-dimensional survey coordinates, while signal_dbm stores the measured received signal strength in dBm. Coordinates must be finite numerical values and signal readings must lie between -120 dBm and 0 dBm.

4.2 Data preparation and validation

The loader checks the required CSV headers and validates individual rows. Empty identifiers, malformed rows, non-finite coordinates and signal readings outside the permitted range are not accepted. A row with a missing coordinate is rejected rather than having its position estimated or imputed. Rejected rows are retained in a separate report with a reason.

4.3 Analysis workflow

1. Read the input survey CSV file.
2. Validate the header and each row; create SurveyPoint objects for accepted measurements.
3. Create tabular representations of the accepted measurements using Pandas.
4. Calculate pairwise distances and grid-cell mean signal values using NumPy.
5. Group readings by location identifier and calculate comparison statistics.
6. Identify readings below the configured weak-signal threshold.
7. Generate charts and export CSV, JSON and Markdown report files.

4.4 Signal interpretation and spatial analysis

The default weak-zone threshold is -67 dBm. A reading below this value is marked as weak for the purpose of this analysis. The application also assigns descriptive signal bands: Excellent (-50 to 0 dBm), Good (-67 to -50 dBm), Fair (-75 to -67 dBm) and Poor (-120 to -75 dBm). These categories are indicative and do not guarantee a particular data rate or user experience.

Grid summaries calculate the mean of the recorded samples that fall inside each cell. A grid cell with no samples remains NaN (unmeasured). The heatmap therefore represents sample-based cell summaries and must not be interpreted as a direct measurement at every point in the cell.

5. Implementation of Major Modules

Module / function	Functionality
SurveyPoint	Immutable data model for a measured signal strength and its coordinates
AnalysisResult	Stores accepted points, rejected rows, comparison tables, weak points and grid results
load_survey_csv()	Reads the CSV file and validates headers and individual records
pairwise_distances()	Computes pairwise Euclidean distances between survey coordinates
grid_summary() / grid_dataframe()	Calculates grid-cell mean signal and creates a tabular grid summary
compare_locations()	Aggregates signal statistics and weak-rate information by location ID
weak_zones()	Selects measurements below the configured signal threshold
Report and plotting functions	Create CSV/JSON/Markdown outputs and Matplotlib visualisations
Command-line workflow	Accepts the input CSV, output directory and grid configuration

5.1 Data model and validation

SurveyPoint is implemented as a frozen dataclass. It stores location_id, x, y, signal_dbm and an optional label. Its validation checks that the identifier is not blank, numerical values are finite and the signal strength lies within the allowed -120 dBm to 0 dBm range. The is_weak property checks the default threshold.

5.2 Output files

Output file	Description
location_comparison.csv	Pandas summary grouped by location ID

weak_zones.csv	Individual measured points below the configured threshold
rejected_rows.csv	Rows excluded from analysis, with rejection reasons
signal_map.png	Coordinate scatter map with weak points highlighted
signal_histogram.png	Distribution of recorded signal readings and threshold
grid_heatmap.png	Visualisation of mean signal strength by grid cell
grid_summary.csv / .npz	Readable grid-cell summary and NumPy data container
summary.json	Machine-readable run metrics and configuration
analysis_report.md	Human-readable report generated from computed results
validated_points.csv	Accepted survey points with signal-band labels

6. Results and Discussion

6.1 Demonstration dataset and descriptive findings

The following values are calculated from the demonstration rows in survey_points.csv in the project repository. They describe this small sample only and should not be treated as a live coverage assessment.

Indicator	Observed value
Input rows	12
Valid records	11
Rejected rows	1
Mean signal strength	-66.91 dBm
Median signal strength	-69.00 dBm
Sample standard deviation	9.86 dB
Minimum / maximum reading	-83 / -52 dBm
Weak threshold	-67 dBm
Weak readings	6 of 11 (54.5%)

Among the 11 accepted readings, 6 fall below the default -67 dBm threshold. This indicates that the demonstration sample includes a mix of stronger and weaker readings across its surveyed coordinates.

6.2 Signal-band distribution

Signal band	Range (dBm)	Number of samples
Excellent	-50 to 0	0
Good	-67 to -50	5
Fair	-75 to -67	4
Poor	-120 to -75	2

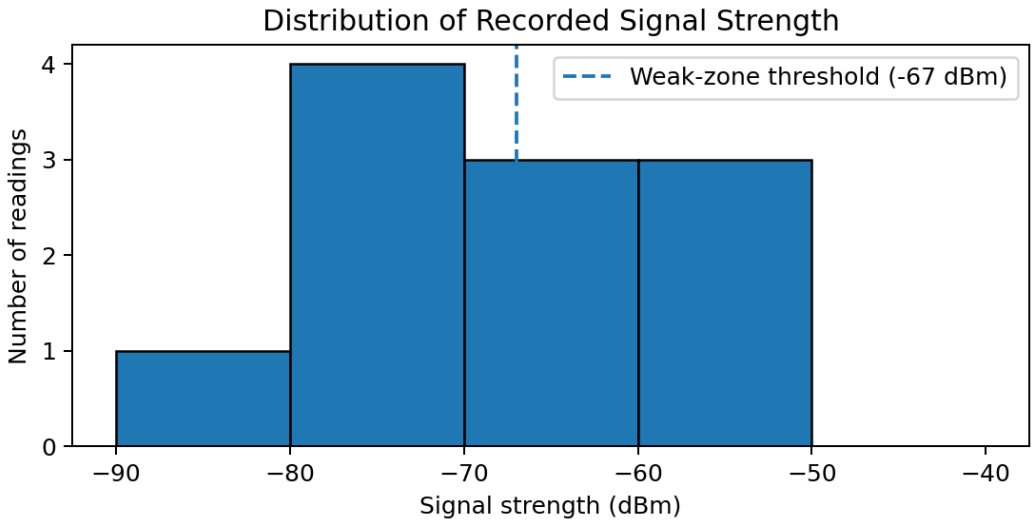
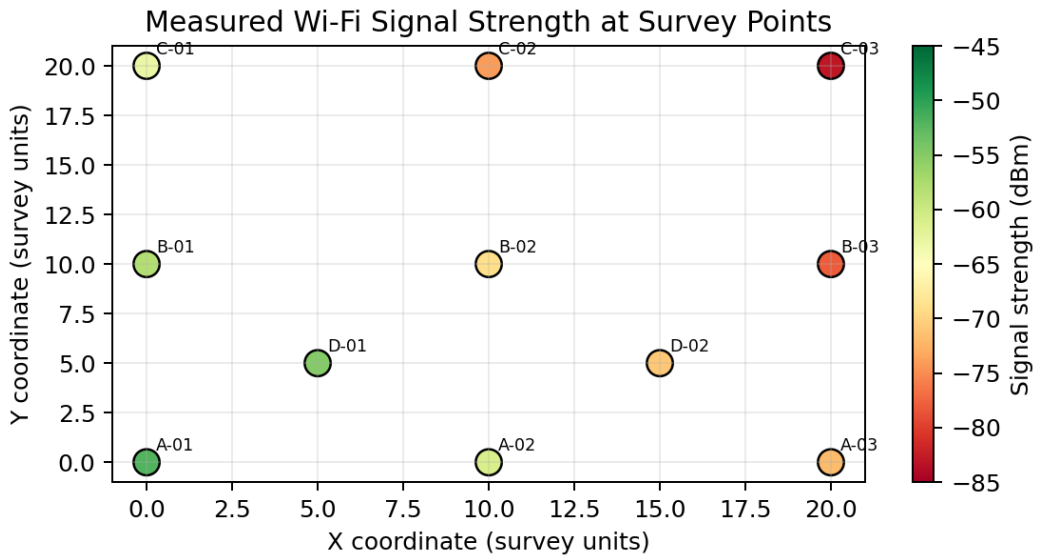


Figure 1. Histogram of recorded signal strengths and the default weak-zone threshold.

6.3 Spatial distribution of signal strength

Figure 2. Measured survey points coloured by recorded signa



l strength.

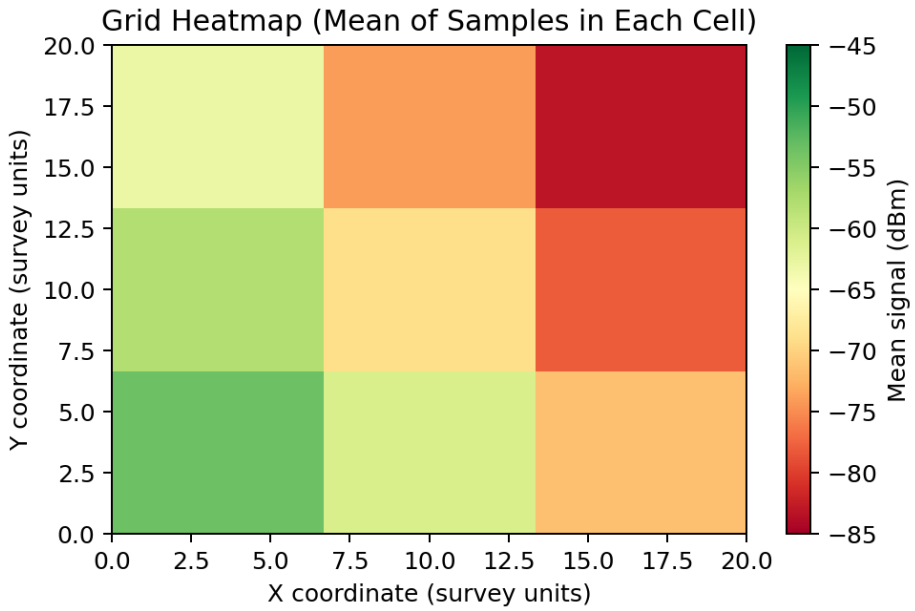


Figure 3. Grid heatmap showing mean signal strength for cells containing samples.

6.4 Location-wise comparison

Location ID	Samples	Mean (dBm)	Min (dBm)	Max (dBm)
A-01	1	-52.00	-52	-52
A-02	1	-61.00	-61	-61
A-03	1	-72.00	-72	-72
B-01	1	-58.00	-58	-58
B-02	1	-69.00	-69	-69
B-03	1	-78.00	-78	-78
C-01	1	-63.00	-63	-63
C-02	1	-74.00	-74	-74
C-03	1	-83.00	-83	-83
D-01	1	-55.00	-55	-55
D-02	1	-71.00	-71	-71

The location-wise table provides a compact comparison of the measurements associated with each location ID. The values are based on the rows in the demonstration CSV; the labels and coordinate layout are illustrative.

6.5 Weak zones and interpretation

Location ID	Area label	X	Y	Signal (dBm)
A-03	East corridor	20	0	-72
B-02	Central room	10	10	-69
B-03	East room	20	10	-78
C-02	Central office	10	20	-74
C-03	Far office	20	20	-83
D-02	Storage	15	5	-71

The weak-zone list identifies individual measurements below -67 dBm. A weak reading may be associated with distance, obstructions, interference, access-point placement or other conditions, but the dataset alone does not establish the cause. Additional site measurements and information about the wireless environment would be required for a detailed diagnosis.

7. Testing and Validation

Test case	Expected behaviour / check	Evidence in project
CSV loading	Accept valid rows and reject missing coordinates	Test expects 11 accepted points and 1 rejected row
Invalid signal strength	Reject a value below -120 dBm	pytest checks ValueError for -121 dBm
CSV header validation	Raise an error when required columns are missing	Test checks missing required column
Malformed row	Reject rows with more fields than the header	Test checks row-level rejection reason
Pairwise distance	Calculate Euclidean distance correctly	Test verifies 3-4-5 distance equals 5
Grid summary	Return cell means and grid edges	Test verifies a one-cell mean of -60 dBm
Location comparison	Aggregate by location and calculate weak rate	Test checks ordering and weak-rate calculation
Custom threshold	Apply the selected threshold consistently	Test verifies weak filtering at -65 dBm

The repository includes pytest tests for input validation, missing coordinates, signal limits, pairwise distances, grid calculations, Pandas comparisons and configurable thresholds. The test file documents the expected behaviour; the report does not claim a separate test execution on the submission computer.

8. Limitations and Future Scope

The current project has the following limitations:

- The included survey_points.csv is a small demonstration dataset and is not a live or representative building survey.
- The tool analyses recorded RSSI values; it does not collect Wi-Fi readings directly from a wireless adapter.
- It does not perform RF propagation simulation or model walls, floor-plan obstructions or access-point antenna patterns.
- It does not perform channel planning or predict throughput.
- Distances use the coordinate units supplied by the survey; the program does not infer real-world units.
- Grid-cell means summarise only the samples inside each cell; empty cells remain unmeasured.

Possible future enhancements include:

- Integration with a Wi-Fi scanning interface to collect measurements during a site survey.
- Import of floor-plan images and coordinate calibration to real distances.
- Support for multiple access points, frequency bands and survey sessions.
- Interactive map navigation and configurable signal thresholds.
- Additional analysis of noise, interference, channel utilisation and signal-to-noise ratio.
- Export of customised PDF reports and comparison of repeated surveys over time.

9. Conclusion

Wireless Coverage Mapper demonstrates a structured Python workflow for analysing recorded Wi-Fi signal strength at two-dimensional survey locations. It combines CSV validation, a reusable SurveyPoint model, NumPy-based spatial calculations, Pandas location comparisons and Matplotlib visualisations. It also separates accepted measurements from rejected rows and distinguishes measured values from grid-cell estimates.

In the repository's demonstration dataset, 11 of 12 rows are accepted and one row is rejected because a coordinate is missing. The default -67 dBm threshold identifies the weak readings in the sample. The

project provides a foundation for systematic wireless site-survey analysis, while its results must be interpreted within the limits of the supplied sample data and the absence of RF propagation or throughput modelling.

10. References

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11. Appendix: CSV Format and Execution

Required CSV headers (exact spelling): location_id, x, y, signal_dbm. The optional field is label.

location_id	x	y	signal_dbm	label
A-01	0	0	-52	Entrance
A-02	10	0	-61	North corridor
A-03	20	0	-72	East corridor
D-03	(blank)	15	-66	Missing coordinate

Install the required packages and run the application from the project directory:

```
python -m pip install -r requirements.txt
python signal_mapper.py survey_points.csv --output-dir analysis_output
pytest
```

The output directory contains the generated comparison tables, rejected-row report, visualisations, grid summaries and analysis report. Grid resolution can be adjusted with the --x-bins and --y-bins options.

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